

Algebra 1
Unit 11
Lesson 1 Homework

Name _____
Date _____
Period _____

1. In the equation $y = a(b)^x$, a represents the ☐ initial amount. ☐ growth factor. and b represents ☐ initial amount ☐ growth factor
2. When b is greater than 1, the equation models ☐ exponential growth. ☐ exponential decay.
3. When b is between 0 and 1, the equation models ☐ exponential growth. ☐ exponential decay.

For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

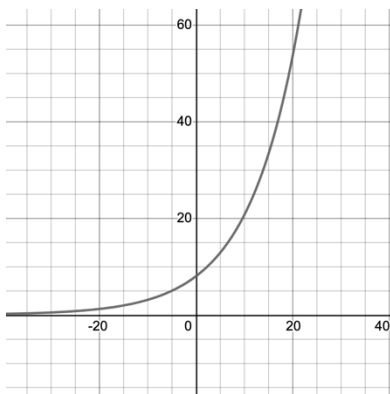
4.

x	y
-1	12
0	4
1	$\frac{4}{3}$
2	$\frac{4}{9}$
3	$\frac{4}{27}$

 represents _____ because...

5. $y = 25(0.79)^x$ represents _____ because...

6.



represents _____
because...

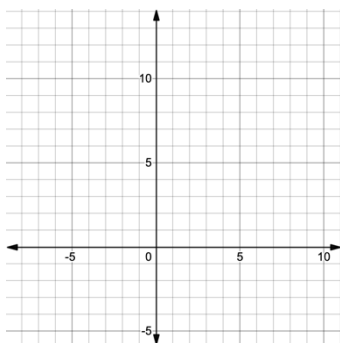
7. $y = 6(1 + 0.05)^x$ represents _____ because...

8.

x	y
2	9
3	13.5
4	20.25
5	30.375

represents _____ because...

9. Sketch a graph that represents exponential growth.



10. Sketch a graph that represents exponential decay.

